Diagnostic Analysis

This presentation provides insights into the utilization of NHS services, trends in missed appointments, and recommendations for improved resource management. The analysis was conducted using Python and NHS data.

Data Overview

- The analysis is based on three datasets:
- 1. actual_duration.csv - Details of appointments and duration
- 2. appointments_regional.csv - Types and status of appointments by region
- 3. national_categories.xlsx - Contextual information and national categorization of appointments.

1. Data Cleaning: Removed unnecessary columns, handled missing values, and ensured data consistency.

2. Data Transformation: Converted date columns to datetime format and aggregated data for analysis.

3. Data Analysis: Explored trends in appointment status, service settings, and regional patterns using visualizations.

Key Insights & Trends

1. Monthly Trends

2. Seasonal Trends

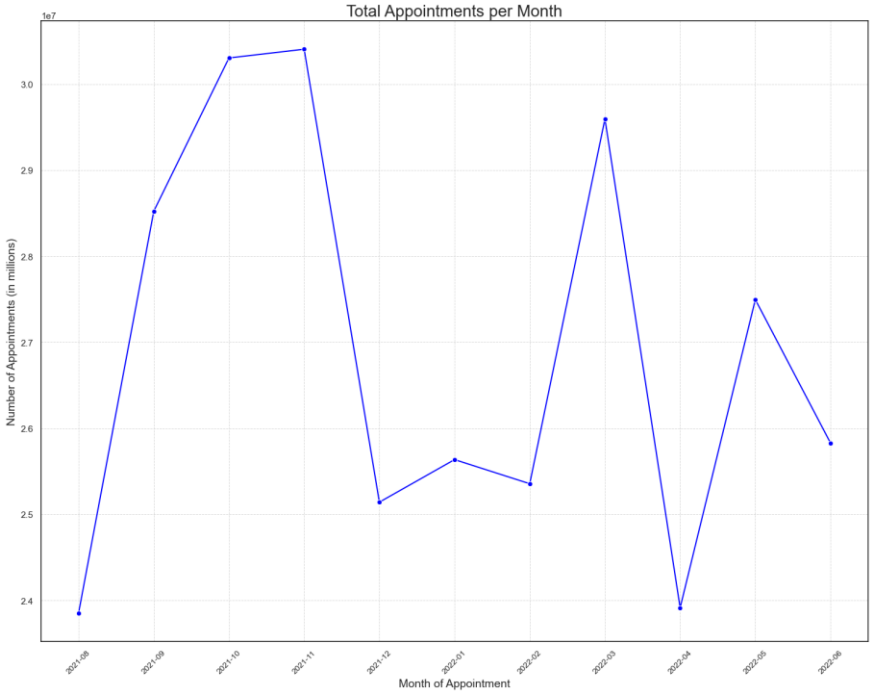
3. Service Settings

4. Missed Appointments

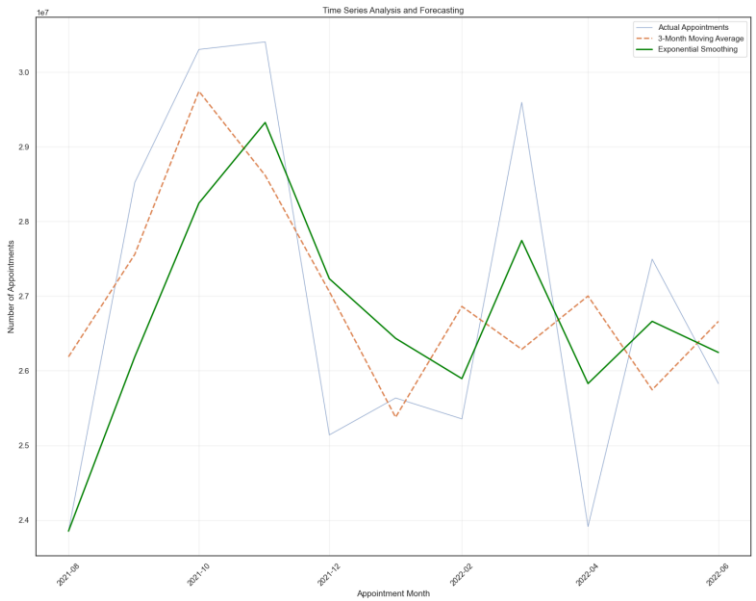
5. Social Media Impact

Notable fluctuations in appointments across months, with peak periods observed in certain months

Monthly Trends



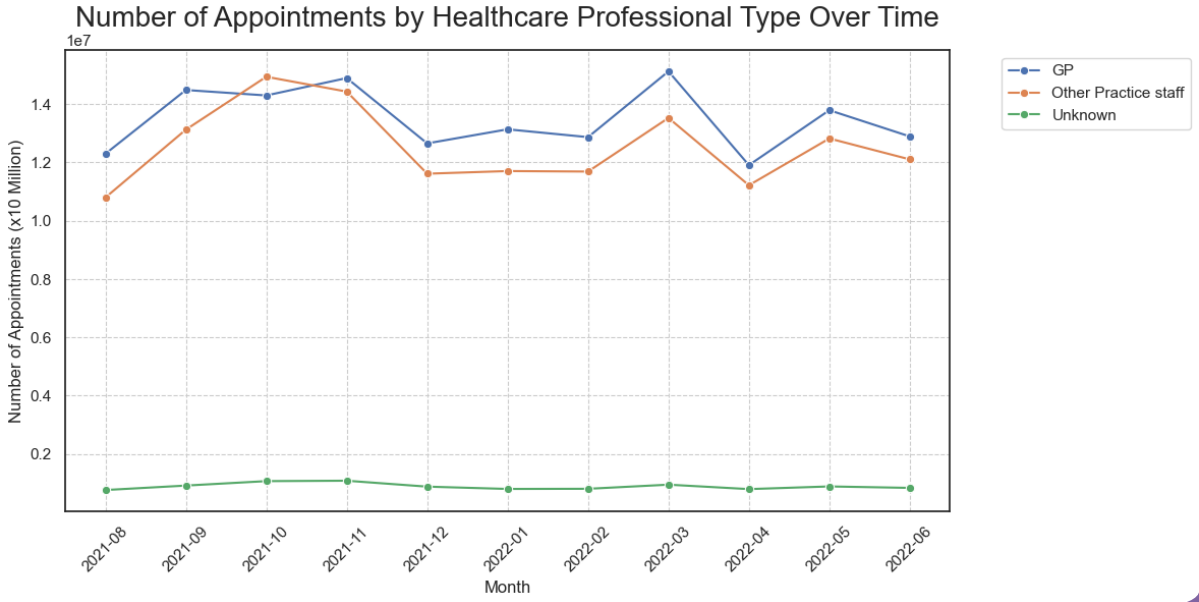
Seasonal Trends



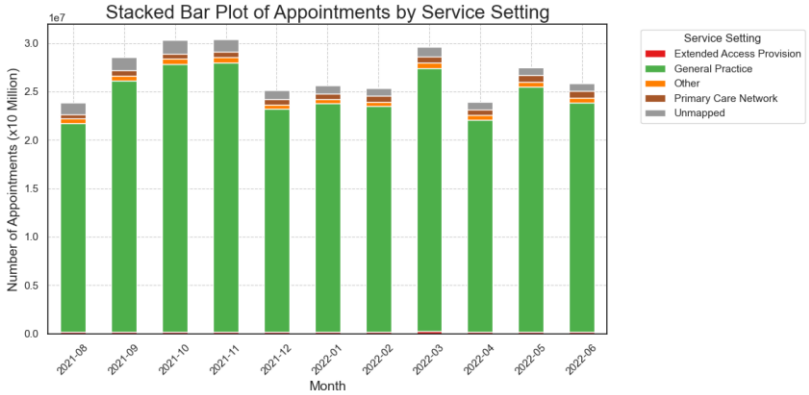
- The 3-Month Moving Average provides a smoother representation of trends by averaging out short-term fluctuations
- The Exponential Smoothing line is more responsive to recent changes than the 3-Month Moving Average, as it gives more weight to recent data points

Seasonal NHS Appointments

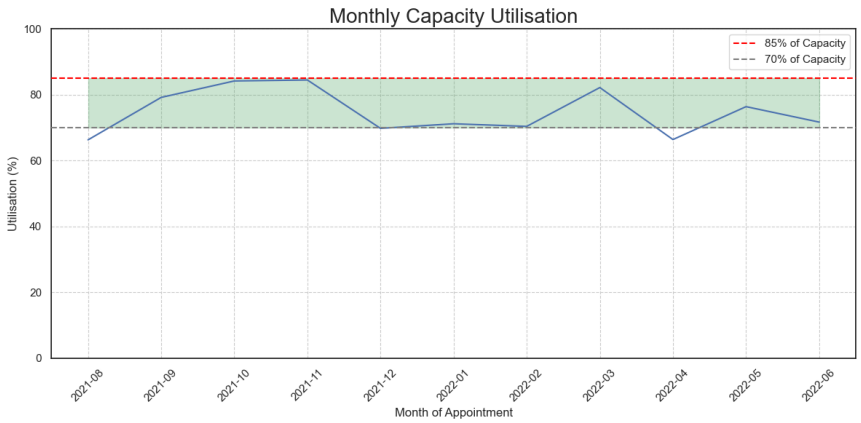
- Different service settings showed varying levels of appointment volumes across seasons.
- But General Practice Dominance consistently the highest number of bookings



Service Settings



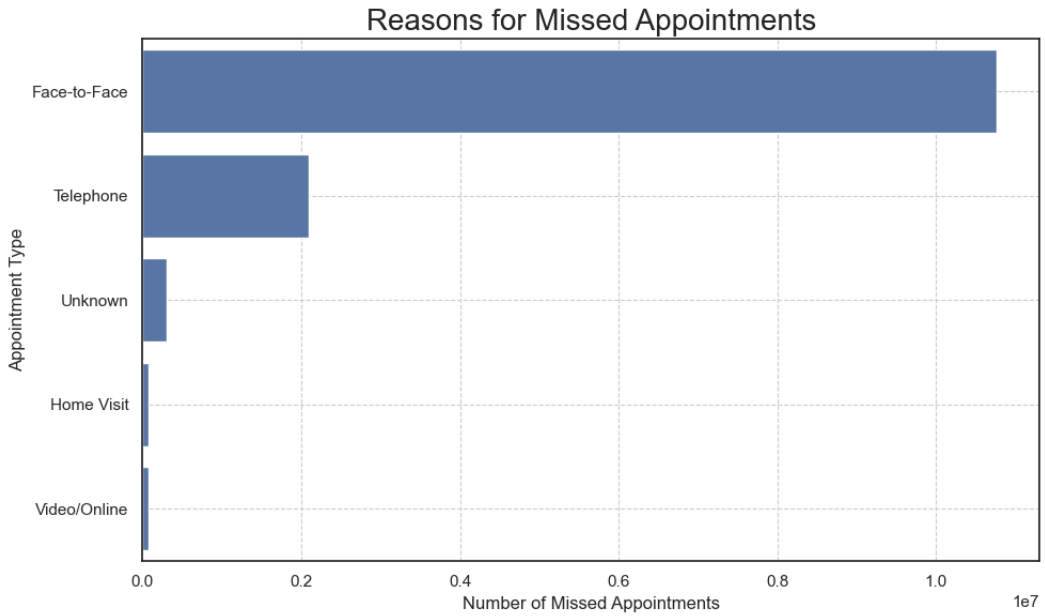
Other service settings, such as Extended Access Provision, Other, and Primary Care Network, each represent a small fraction of the total appointments.



NHS Capacity Utilisation

- Capacity utilisation fluctuated between 66.3% in August 2021 and a peak of 84.5% in November 2021.
- Months with lower utilisation suggest some unused capacity, which might reflect seasonally lower demand.

Missed Appointments



- Face-to-Face representing the highest volume of missed appointments
- Video/Online appointments a close second

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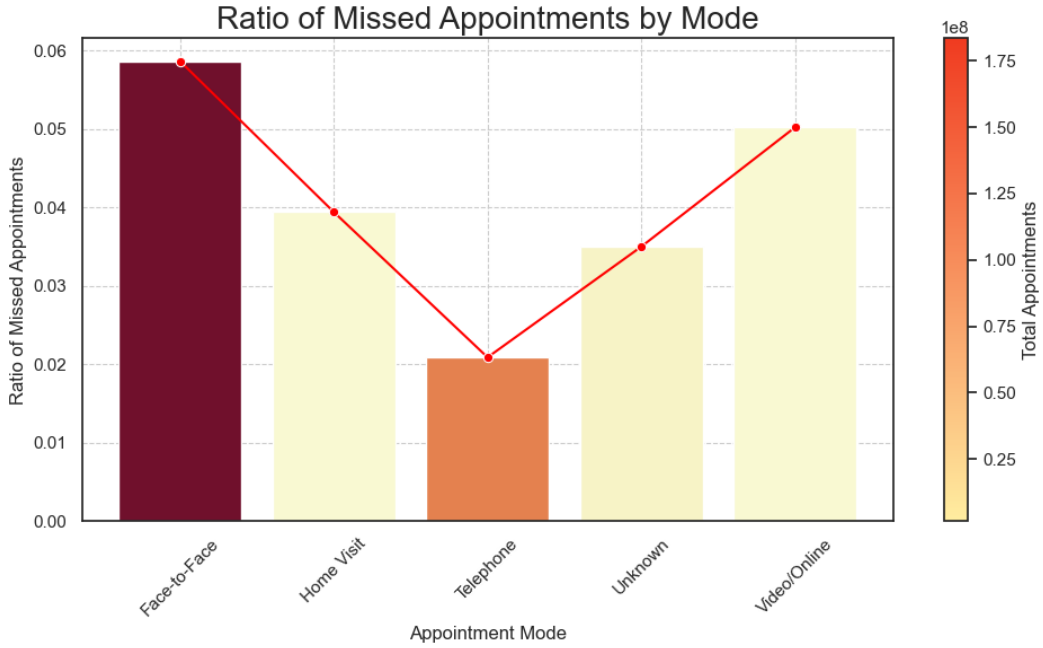

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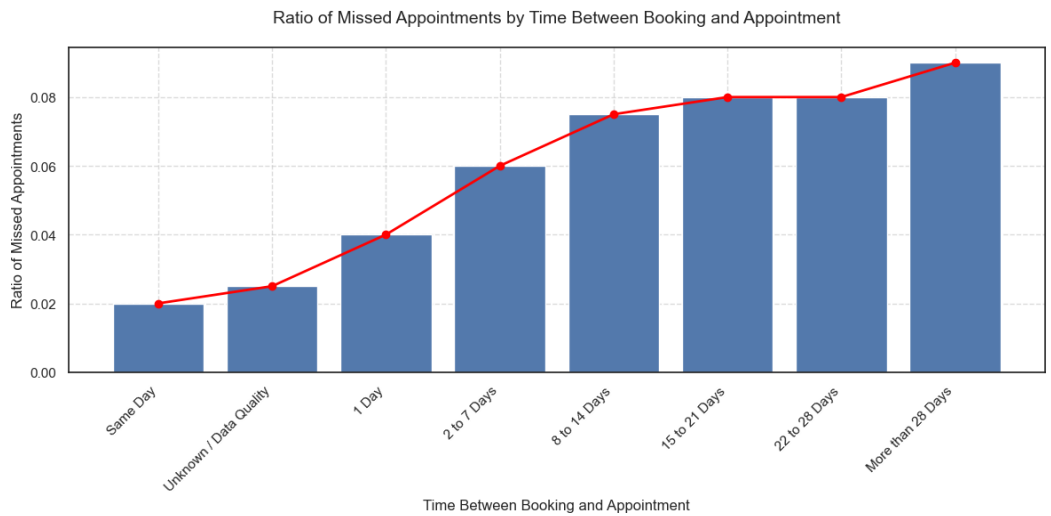


- Same day bookings are less likely to be missed
- As the appointment times get longer the more they are missed

Recommendations

1. Improve communication channels to reduce missed appointments, particularly during peak periods.

Recommendations

2. Reallocate resources to regions with higher demand during peak seasons.

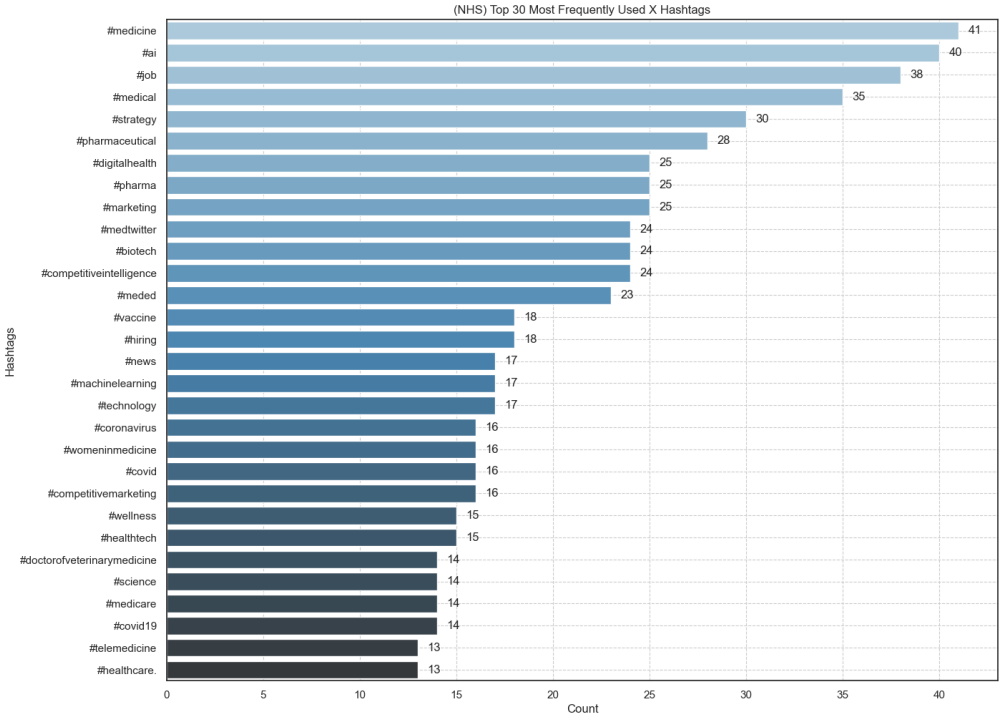
Recommendations

3. Enhance data collection methods to ensure consistent reporting of appointment status and durations.

Social Media Impact

An opportunity for the NHS to address public concerns and improve patient communication

With the ability to track the use of key words and trends on Twitter(X). It is a great opportunity to get ahead of the chaos



Conclusion

The analysis highlights key trends in NHS appointment utilization and missed appointments. By understanding these patterns, the NHS can better allocate resources, improve patient outcomes, and reduce unnecessary costs associated with missed appointments.